

26 Expected Values of Linear Combinations of Random Variables

A **linear rescaling** of a random variable X is of the form $aX + b$ where a and b are non-random constants. A **linear combination** of two random variables X and Y is of the form $aX + bY$ where a and b are non-random constants.

Properties of expected value

- Expected value and linear rescaling: If X is a random variable and a, b are non-random constants then

$$E(aX + b) = aE(X) + b$$

- Linearity of expected value: For *any* two random variables X and Y ,

$$E(X + Y) = E(X) + E(Y)$$

- That is, the expected value of the sum is the sum of expected values, regardless of how the random variables are related.
- Therefore, you only need to know the marginal distributions of X and Y to find the expected value of their sum. (But keep in mind that the *distribution* of $X + Y$ will depend on the joint distribution of X and Y .)
- Combining properties of linear rescaling with linearity of expected value yields the expected value of a linear combination, which extends naturally to more than two random variables.

$$E(aX + bY + c) = aE(X) + bE(Y) + c$$

Properties of variance

- Variance and linear rescaling: If X is a random variable and a, b are non-random constants then

$$SD(aX + b) = |a|SD(X)$$

$$Var(aX + b) = a^2Var(X)$$

- Variance of sums and differences of random variables

$$Var(X + Y) = Var(X) + Var(Y) + 2Cov(X, Y)$$

$$Var(X - Y) = Var(X) + Var(Y) - 2Cov(X, Y)$$

- The variance of the sum (or difference) is the sum of the variances if and only if X and Y are uncorrelated.

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) \quad \text{if } X, Y \text{ are uncorrelated}$$

$$\text{Var}(X - Y) = \text{Var}(X) + \text{Var}(Y) \quad \text{if } X, Y \text{ are uncorrelated}$$

- If a, b, c are non-random constants and X and Y are random variables then

$$\text{Var}(aX + bY + c) = a^2\text{Var}(X) + b^2\text{Var}(Y) + 2ab\text{Cov}(X, Y)$$

Example 26.1. Recall the SAT score Application assignment.

Let M be the Math score and R be the Reading (and Writing) score of a randomly selected SAT taker. We'll assume (M, R) pairs follow a *Bivariate Normal distribution*.

- Math scores (M) have mean 500 and standard deviation 140
- Reading scores (R) have mean 520 and standard deviation 110

For each of the following scenarios, compute

- $E(M + R)$
- $\text{Var}(M + R)$
- $\text{SD}(M + R)$

1. $\text{Corr}(M, R) = 0$

2. $\text{Corr}(M, R) = 0.6$

3. $\text{Corr}(M, R) = -0.9$

4. In which of these scenarios is $\text{SD}(M + R)$ largest? Smallest? Can you explain why intuitively?

Example 26.2. Recall the SAT score Application assignment.

Let M be the Math score and R be the Reading (and Writing) score of a randomly selected SAT taker. We'll assume (M, R) pairs follow a *Bivariate Normal distribution*.

- Math scores (M) have mean 500 and standard deviation 140
- Reading scores (R) have mean 520 and standard deviation 110

For each of the following scenarios, compute

- $E(M - R)$
- $\text{Var}(M - R)$
- $\text{SD}(M - R)$

1. $\text{Corr}(M, R) = 0$

2. $\text{Corr}(M, R) = 0.6$

3. $\text{Corr}(M, R) = -0.9$

4. In which of these scenarios is $\text{SD}(M - R)$ largest? Smallest? Can you explain why intuitively?

- Two continuous random variables X and Y have a **Bivariate Normal** distribution with parameters $\mu_X, \mu_Y, \sigma_X > 0, \sigma_Y > 0$, and $-1 < \rho < 1$ if the joint pdf is

$$f_{X,Y}(x,y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2(1-\rho^2)} \left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2 - 2\rho\left(\frac{x-\mu_X}{\sigma_X}\right)\left(\frac{y-\mu_Y}{\sigma_Y}\right) \right]\right)$$

- It can be shown that if the pair (X, Y) has a BivariateNormal($\mu_X, \mu_Y, \sigma_X, \sigma_Y, \rho$) distribution

$$E(X) = \mu_X$$

$$E(Y) = \mu_Y$$

$$SD(X) = \sigma_X$$

$$SD(Y) = \sigma_Y$$

$$\text{Corr}(X, Y) = \rho$$

- A Bivariate Normal Density has elliptical contours. For each height $c > 0$ the set $\{(x, y) : f_{X,Y}(x, y) = c\}$ is an ellipse. The density decreases as (x, y) moves away from (μ_X, μ_Y) , most steeply along the minor axis of the ellipse, and least steeply along the major of the ellipse.
- A scatterplot of (x, y) pairs generated from a Bivariate Normal distribution will have a rough linear association and the cloud of points will resemble an ellipse.
- If X and Y have a Bivariate Normal distribution, then the marginal distributions are also Normal: X has a Normal(μ_X, σ_X) distribution and Y has a Normal(μ_Y, σ_Y).
- If X and Y have a Bivariate Normal distribution and $\text{Corr}(X, Y) = 0$ then X and Y are independent. (Remember, in general it is possible to have situations where the correlation is 0 but the random variables are not independent.)
- **X and Y have a Bivariate Normal distribution if and only if every linear combination of X and Y has a Normal distribution.** That is, X and Y have a Bivariate Normal distribution if and only if $aX + bY$ has a Normal distribution for all a, b .

Example 26.3. Let M be the Math score and R be the Reading (and Writing) score of a randomly selected SAT taker. We'll assume (M, R) pairs follow a *Bivariate Normal distribution*.

- Math scores (M) have mean 500 and standard deviation 140
- Reading scores (R) have mean 520 and standard deviation 110
- The correlation is 0.6

1. Compute the probability that a student has a total score above 1400.
2. Compute the probability that a student has a higher Math than Reading score.

Properties of covariance

$$\text{Cov}(X, X) = \text{Var}(X)$$

$$\text{Cov}(X, Y) = \text{Cov}(Y, X)$$

$$\text{Cov}(X, c) = 0$$

$$\text{Cov}(aX + b, cY + d) = ac\text{Cov}(X, Y)$$

$$\text{Cov}(X + Y, U + V) = \text{Cov}(X, U) + \text{Cov}(X, V) + \text{Cov}(Y, U) + \text{Cov}(Y, V)$$

Example 26.4. Let X be the number of two-point field goals a basketball player makes in a game, Y the number of three point field goals made, and Z the number of free throws made (worth one point each). Assume X, Y, Z have standard deviations of 2.5, 3.7, 1.8, respectively, and $\text{Corr}(X, Y) = 0.1$, $\text{Corr}(X, Z) = 0.3$, $\text{Corr}(Y, Z) = -0.5$.

1. Find the standard deviation of the number of fields goals in a game (not including free throws)
2. Find the standard deviation of total points scored on fields goals in a game (not including free throws)

3. Find the standard deviation of total points scored in a game.